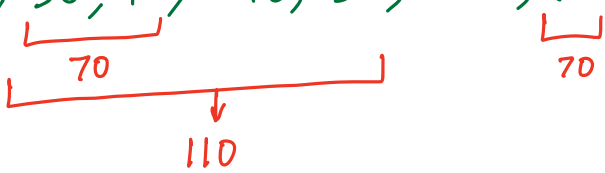


About Myself

- SWE III @ Google
- 3 times ACM-ICPC Regionals
- 3 times in top 2000 in FB/Meta Hacker Cup
- Master at Codeforces
- 6 star at Codechef
- BTech Hons. in CSE @ IIT Bhilai
- Former member of National Blockchain Project, IIT-K
- Author @ CCGuid '21

Q1) Given an $[n]$, find the max subarray sum.

$$\text{arr} \rightarrow \{-20, 30, 40, -10, 50, -60, 70\}$$



$$\{4, 5, 2, 1, 6\} \rightarrow \text{ans} = 18$$

$$\{-4, -3, -6, -9, -2\} \rightarrow \text{ans} = -2$$

$$\{-20, 30, 40, -10, 50, -60, 70\}$$

start

$$\boxed{i=0}$$

$$j=0 \rightarrow \text{sum} = -20$$

$\downarrow +30$

$$j=1 \rightarrow \text{sum} = 10$$

$\downarrow +40$

$$j=2 \rightarrow \text{sum} = 50$$

$\downarrow +(-10)$

$$j=3 \rightarrow$$

40

⋮

$$[i, j] = [0, 2] \xrightarrow{\text{sum}} 50.$$

$$[0, 3] \xrightarrow{\text{sum}} 40.$$

$$\boxed{i=1}$$

$$j=1 \rightarrow \text{sum} = 30$$

$\downarrow +40$

$$j=2 \rightarrow \text{sum} = 70$$

⋮

ans = $-\infty$

for $i \rightarrow 0$ to $n-1$ {

sum = 0

for $j \rightarrow i$ to $n-1$ {

sum += arr[j]

if (sum > ans)

ans = sum

}

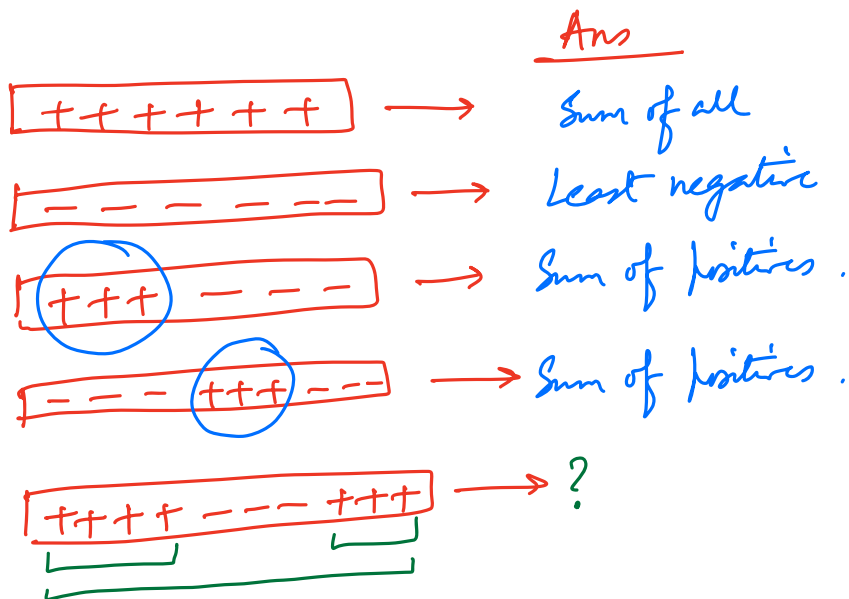
}

return ans

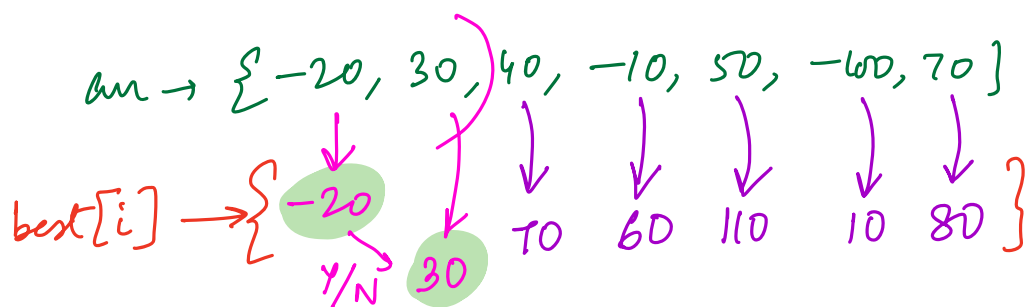
CARRY FORWARD

$O(n^2)$ TC

$O(1)$ SC



$best_i[i] = \text{max sum of subarray ending at } i.$

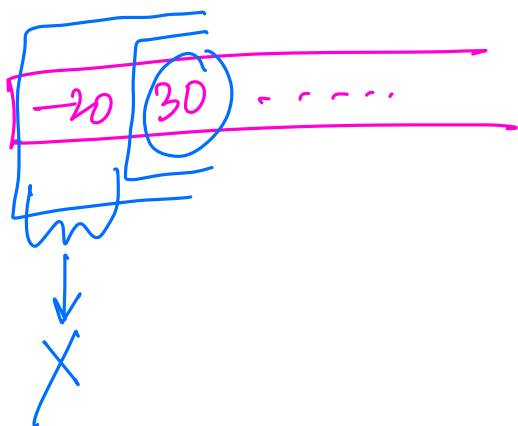


$\text{max}(30, 30 + (-20))$

$\text{max}(-10, -10 + 70)$

40

$40 + \underbrace{\text{best before}}_{30} \rightarrow 70$



$ans = -\infty$

$sum = 0$

for $i \rightarrow 0$ to $n-1$ {

$sum = \text{max}(arr[i], arr[i] + sum)$

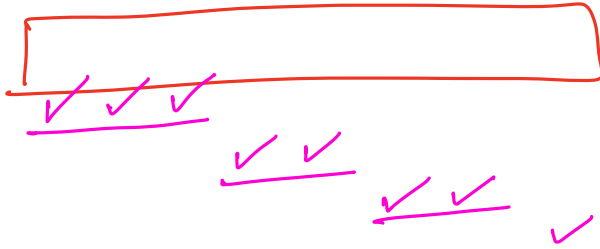
$ans = \text{max}(sum, ans)$

}

Kadane's
Algr

$sum \rightarrow$ Max sum of any subarray ending at i

$ans \rightarrow$ Best subarray sum in $arr[0 \dots i]$.



$ans = -\infty$

$s = -1, e = -1$

$sum = 0$

$temp-s = 0$

for $i \rightarrow 0$ to $n-1$ {
 if ($sum \leq 0$) {

$sum = 0$

$temp-s = i$

}

$sum += arr[i]$

if ($sum > ans$) {

$s = temp-s$

$e = i$

$ans = sum$

}

}

$\{-2, 3, 4, -1, 5, -10, 7\}$

i	sum	$temp-s$	s	e	ans
	0	0	-1	-1	$-\infty$
0	-2	0	0	0	-2
1	3	1	1	1	3
2	7		1	2	7
3	6				7
4	11		1	4	11
:					

$O(n)$ T.C.

$O(1)$ S.C.

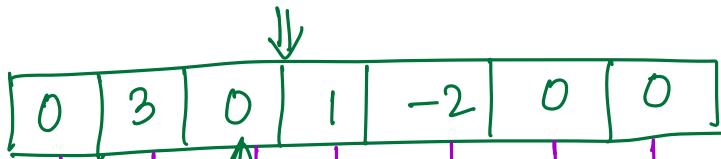
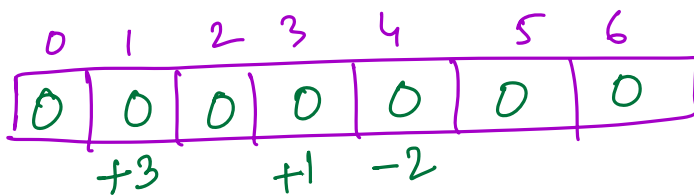
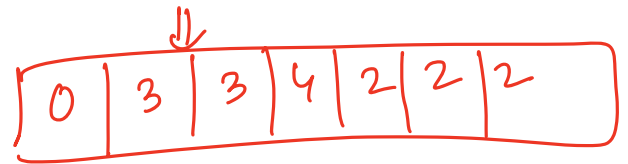
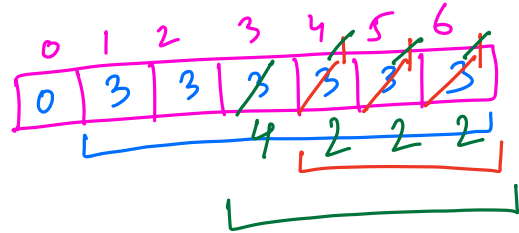
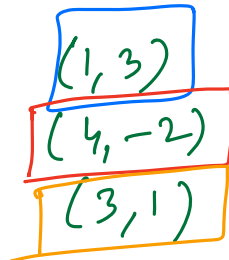
[Break till 10:19 PM]

Q2) Given $A[n]$, where each element is 0. After the following queries, return the final array.

Each query $\vdash (i, x) \Rightarrow$ Add x to all the elements from index i to $n-1$.

eg:- $n=7$

queries:-



Running
Prefix
sum



$A[n] = \{0, 0, \dots, 0\}$

```
for  $i \rightarrow 0$  to  $q-1$  {  
     $ind = queries[i][0]$   
     $x = queries[i][1]$   
     $A[ind] += x$   
}
```

```
for  $i \rightarrow 1$  to  $n-1$  {  
     $A[i] = A[i-1] + A[i]$   
}  
return A
```

$O(n+q)$ TC.
 $O(1)$ S.C

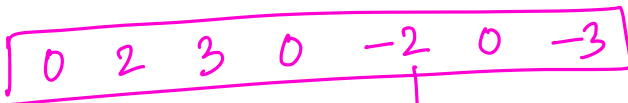
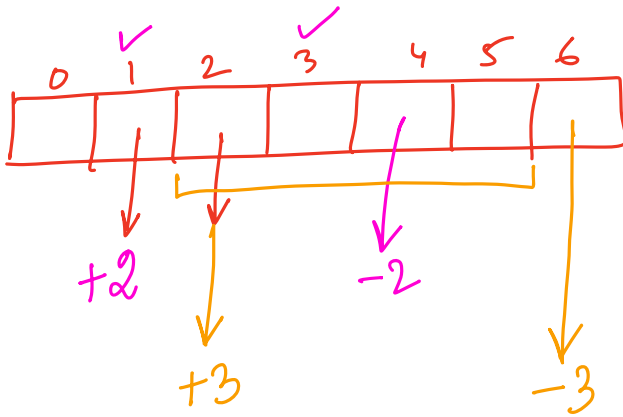
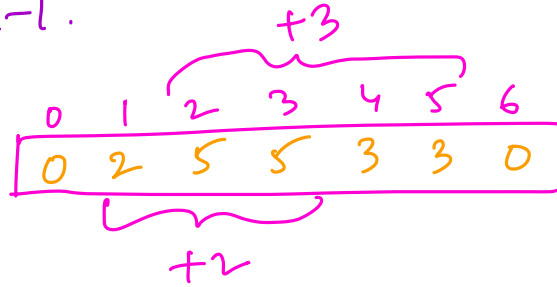
Q3) Given $A[n]$, where each element is 0. After the following queries, return the final array.

Each query $:- (i, j, x) \Rightarrow$ Add x to all the elements from index i to j .
 $0 \leq i \leq j \leq n-1$.

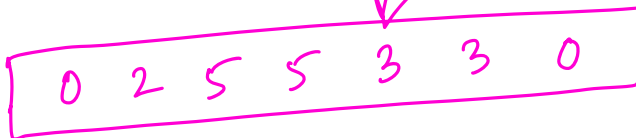
$n=7$

$q_1 \rightarrow (1, 3, 2)$

$q_2 \rightarrow (2, 5, 3)$



↓ Prefix Sum



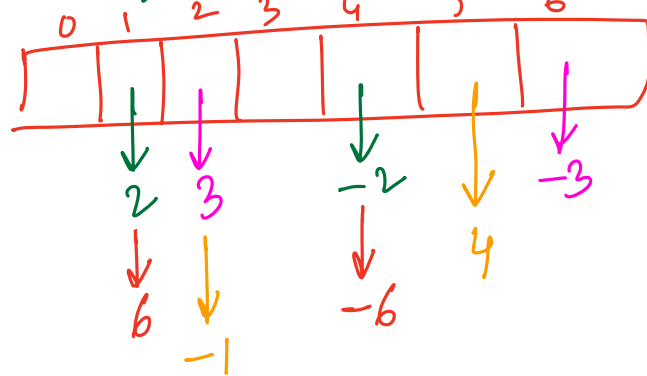
$n=7$

$q_1 \rightarrow (1, 3, 2) \checkmark$

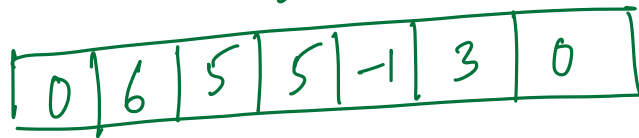
$q_2 \rightarrow (2, 5, 3) \checkmark$

$q_3 \rightarrow (1, 3, 4) \checkmark$

$q_4 \rightarrow (2, 4, -4) \checkmark$



↓ PS



$A[n] = \{0, 0, \dots, 0\}$

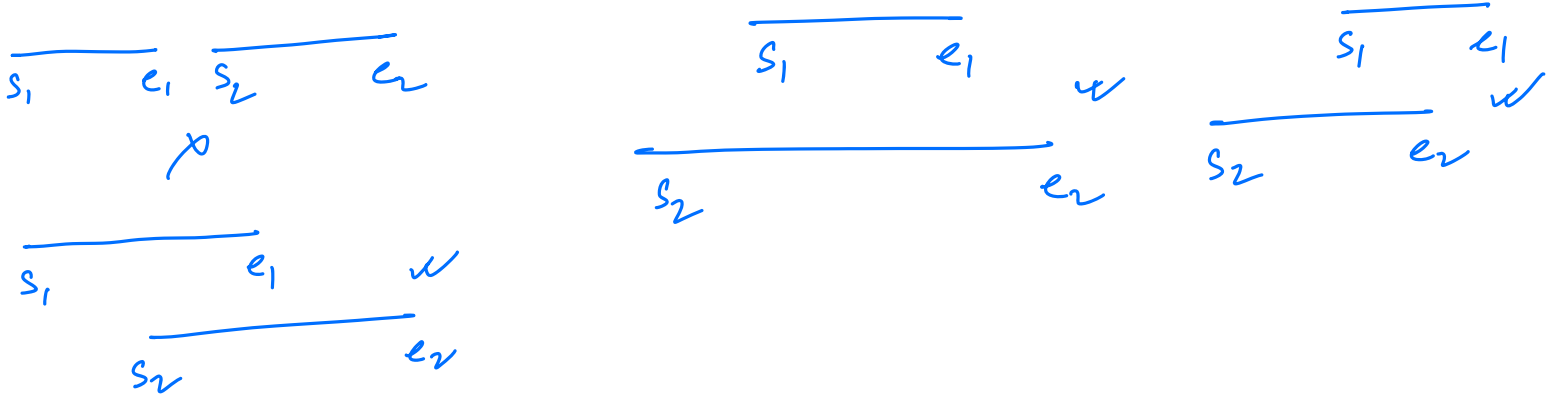
```
for  $i \rightarrow 0$  to  $q-1$  {  
     $s = \text{queries}[i][0]$   
     $e = \text{queries}[i][1]$   
     $x = \text{queries}[i][2]$   
     $A[s] += x$   
    if ( $e < n-1$ )  
         $A[e+1] -= x$ 
```

$O(n+q)$ TC.

$O(1)$ S.C

```
}  
for  $i \rightarrow 1$  to  $n-1$  {  
     $A[i] = A[i-1] + A[i]$   
}  
return A
```

Given two intervals (s_1, e_1) and (s_2, e_2) , find if they overlap.

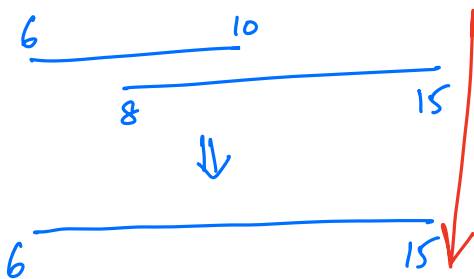


Non-overlapping $\rightarrow$ One starts after the other ends

$$(s_1 > e_2 \text{ OR } s_2 > e_1)$$

Overlapping $\rightarrow (s_1 \leq e_2) \text{ AND } (s_2 \leq e_1)$

$$\max(s_1, s_2) \leq \min(e_1, e_2)$$

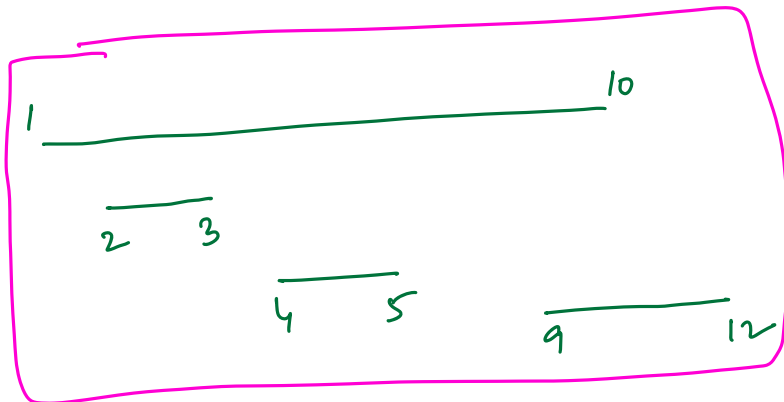
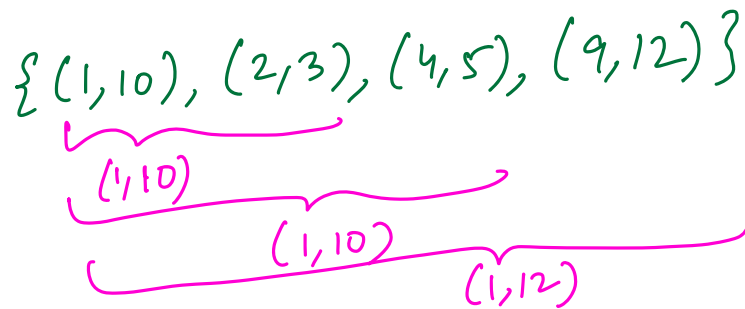
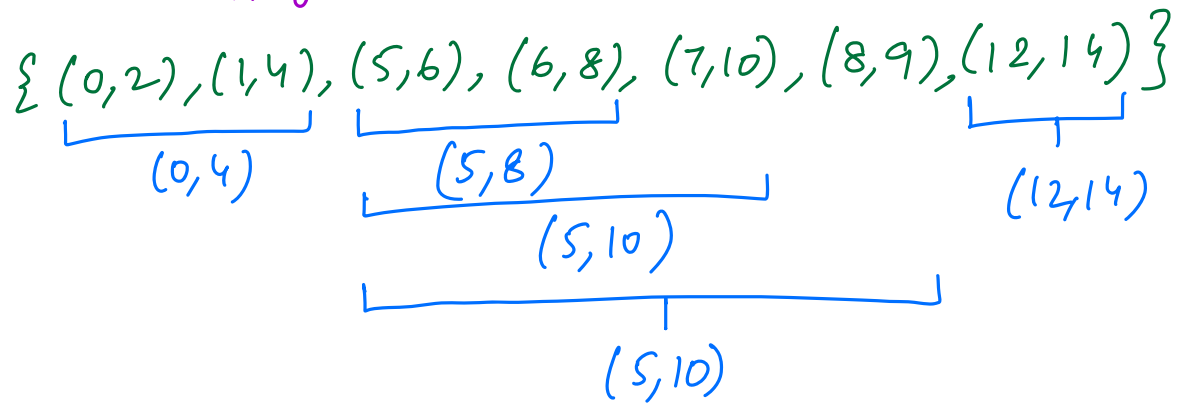


Merge the two intervals

Merged interval $\rightarrow [\min(s_1, s_2), \max(e_1, e_2)]$

Q4) Given n intervals of the format $(start, end)$, sorted in increasing order of the start time.

Merge the overlapping intervals and return the final set of non-overlapping intervals.



$currS = A[0][0]$

$currE = A[0][1]$

$findSet = \{ \}$

for $i \rightarrow 1$ to $n-1$ {

if ($A[i][0] \leq currE$) {

$currS = \min(currS, A[i][0])$

$currE = \max(currE, A[i][1])$

}

else {

$findSet.add(\{currS, currE\})$

$currS = A[i][0]$

$currE = A[i][1]$

}

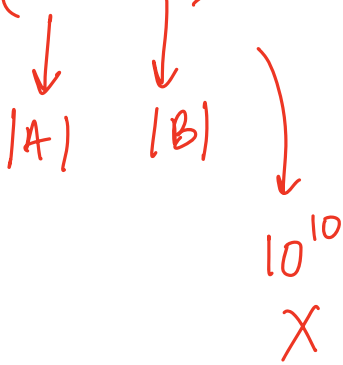
}
 $findSet.add(\{currS, currE\})$

return findSet

$O(n)$ T.C.

$O(1)$ S.C

$$O(n * m)$$

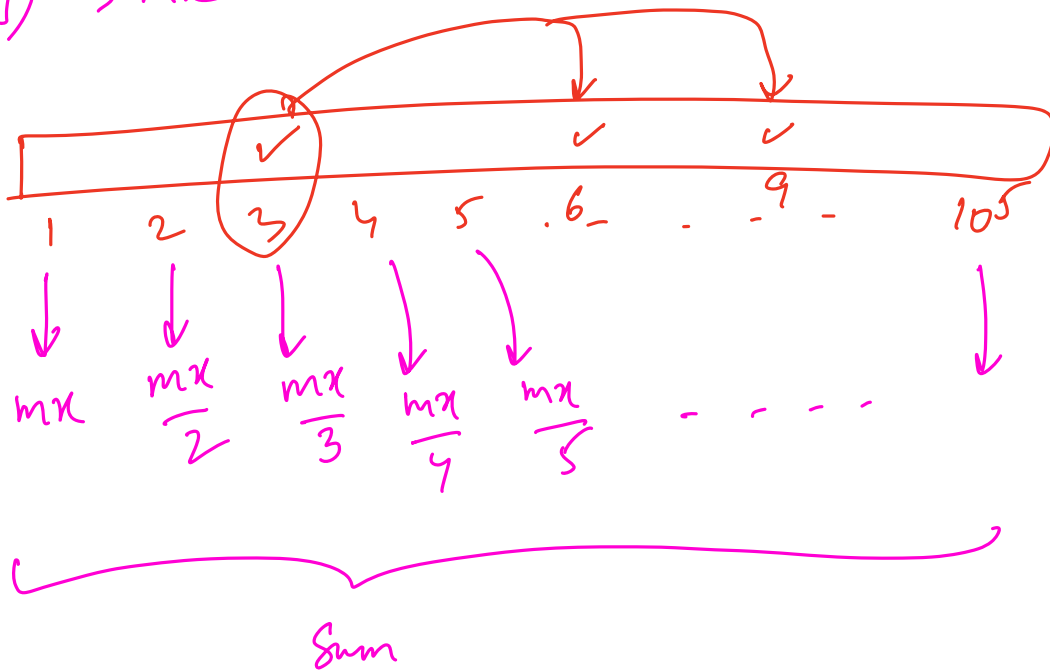


$$O(n * \sqrt{\max(A[i])} + m)$$

$n \rightarrow 10^5$
 $1 \leq A[i] \leq 10^5$
 $m \rightarrow 10^5$

$\boxed{10^5 * 300}$
 ✓

$$\max(A[i]) \rightarrow mx$$



$$= mx \left(\frac{1}{1} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{mx} \right)$$

$$\approx O(mx * \log(mx))$$

$$TC \rightarrow O(\max(A[i]) * \log(\max(A[i])) + \log(A) + \log(B))$$

$n \rightarrow 10^5$
 $1 \leq A[i] \leq 10^5$
 $m \rightarrow 10^5$

$10^5 * 20 + 10^5 + 10^5$
 $= 2 * 10^6$

$$\begin{array}{|c|} \hline 2 \\ \hline \end{array}$$

Ans

3

8

12

↓

$$\begin{array}{cccccc|c}
 2 & -3 & 3 & 5 & -2 & 6 & -100 & 5 \\
 \hline
 3 & 8 & 6 & 12 & -88 & & & 5
 \end{array}$$